E-JUST

الجامعة المصرية اليابانية للعلوم و التكنولوجيا
エジプト日本科学技術大学
EGYPT-JAPAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Electronics, Communications, and Computer Engineering
Department of Electronics, and Communications Engineering

Final Project Report

AI Enabled Smart Nutrient Balancing System for Controlled Hydroponic Systems

Presented to:

Prof. Adel Bedair

Eng. Mohamed Ashraf

Eng. Mahmoud Al Azhary

Team Members:

1. **Kareem Galal Hassan** (120230108)
2. **Amr Saad Abdelgabder** (120230132)
3. **Omar Walid Elsayed** (120230143)

Abstract

This report outlines the development of an AI-Enabled Smart Nutrient Balancing System to better manage Controlled Environment Agriculture (CEA) with a focus on converting the traditional (reactive) management of Hydroponic Systems into a pro-active method by removing uncertainty through forecasting of conditions in real-time using IoT technologies.

The management of Hydroponic Systems requires continuous and precise (to prevent loss of crops) control of physical and chemical variables; however, the methods currently used to manage Hydroponic Systems are through the use of threshold-based controllers, which typically only respond reactively after exceeding pre-determined thresholds and have a high delay time (temporal latency).

To solve this problem, The Development of an AI-Enabled Smart Nutrient Balancing Language and Control System was accomplished by integrating an IoT Environmental Sensing Layer onto an ESP32 Microcontroller with a Hybrid Convolutional Neural Network and Long Short-Term Memory Prepared Predictive Model which would forecast conditions before they occur rather than responding after the fact.

This predictive capability will also allow for chemical actuation shortly (15 minutes) prior to needing to be made to an environment rather than relying on operators to monitor data and then react to exceedances for the purpose of maintaining optimum growth conditions for plants grown in Hydroponic Systems.

Table of contents

Abstract 2

1. Introduction 5

1.1 Background 5

1.2 Problem Statement 5

1.3 Proposed Solution 6

1.4 Project Timeline 7

1.5 Report Organization 7

2. Related Works 8

2.1 Hydroponic Methods and Sensor Integration 8

2.2 IoT and Machine Learning in Agriculture 9

2.3 Comparison of Prior Solutions 11

3. System Architecture and Methodology 12

3.1 Overall System Architecture 12

3.2 Hardware Design and Environmental Sensing 14

3.3 Software Components and Data Flow 16

3.3.1 Data Flow and Communication 17

3.4 Edge Firmware Implementation 19

3.4.1 System Overview and Architecture 19

3.4.2 Dependencies and Pin Configuration 19

3.4.3. Initialization and Calibration Phase (setup) 19

3.4.4. Non-Blocking Execution Model (loop) 20

3.4.5. Sensor Data Processing Logic 20

3.4.6. Cloud Telemetry and Data Logging 21

4. Artificial Intelligence Model 22

4.1 Model Background & Challenges 22

4.2 Dataset Selection and Characteristics 24
4.3 Artificial Intelligence Algorithm Implementation 24
4.4 Model Evaluation and Results 26
5. Experimental Setup, System Validation, and Results 28
5.1 Experimental Setup and Sensor Calibration 28
5.2 System Validation and Analysis 30
5.3 Results and Overall System Accuracy 31
5.4 Interpretation of Results 32
5.5 Discussion on Real-Time Performance and Privacy 34
5.6 Application Context and Deployment 34
6. Conclusion 37
7. References 38
8. Appendix 38

1. Introduction

1.1 Background

The transition from conventional soil-based agriculture to Controlled Environment Agriculture (CEA) has primarily arisen due to increasing needs for higher crop yields, conserving resources, and providing resilience against unpredictable climate patterns. Hydroponics is an extremely effective method of growing plants without soil in a controlled environment through tightly controlled mineral nutrient solutions dissolved into a water solvent. Hydroponics can be implemented as wick systems, deep water cultures, nutrient film techniques (NFT), ebb and flow systems, and aeroponic systems. Water is always the sole and absolute medium for nutrient delivery to the plant root zone and for the exchange of oxygen in the plant root zone.

For a typical Electronics and Communication Engineer, a hydroponic greenhouse does not just function as a conventional farm but rather exists as a very complex multi-variable dynamic control system. The desired output of this system is represented by the biological plants that grow, whereas the inputs include all freshwater parameters and all chemical/environmental parameters such as potential hydrogen, total dissolved solids, ambient humidity, and fluid temperature. This entire ecosystem is dependent upon the consistent and precise balancing of these variables. On the other hand, traditional soil-based farming systems possess an immense amount of naturally occurring buffering capacity that is able to neutralize acidic elevations or store excess salts, while hydroponic systems provide no natural buffering capability. Consequently, real-time monitoring and rapid, automated responses are not just beneficial, but absolutely critical to prevent sudden chemical lockout, irreversible cellular damage, or root death.

1.2 Problem Statement

The management of Controlled Environment Agriculture (CEA), specifically hydroponics, is hindered by the stochastic nature of biological variables and the limitations of traditional monitoring frameworks. Current conventional methods rely on manual point-in-time sampling, which creates high temporal latency. In a soil-less medium, the lack of a natural buffer means undetected fluctuations in pH or nutrient concentration can lead to rapid systemic crop failure before corrective action can be taken.

Furthermore, a hydroponic system is a complex, multivariate environment where variables such as light intensity and air temperature drive the rate of transpiration, which in turn non-

linearly alter water levels and nutrient density. Standard threshold-based controllers fail to account for these interdependencies, often leading to oscillatory instability through over-correction. Existing automated systems are primarily reactive, triggering actuators only after a parameter has breached a pre-defined limit. This necessitates an intelligent predictive layer capable of performing trend analysis to forecast suboptimal states before they occur, transitioning the system from reactive maintenance to proactive optimization.

1.3 Proposed Solution

This project will develop an AI-Enabled Smart Nutrient Balancing System to solve the critical problems with time delay and instability of traditional, reactive hydroponic systems (reactive management). This new approach to hydroponic management will change the architecture of control systems from reactive maintenance to a proactive optimization model and enable sub-optimal conditions to be anticipated and addressed before they happen.

Figure 1: Illustrative design for the proposed prototype and system

1.4 Project Timeline

The development and execution of the AI-Enabled Smart Nutrient Balancing System followed a comprehensive 15-week schedule, divided into specific technical phases to ensure systematic progression from conceptualization to final deployment, as shown in the following Gantt chart:

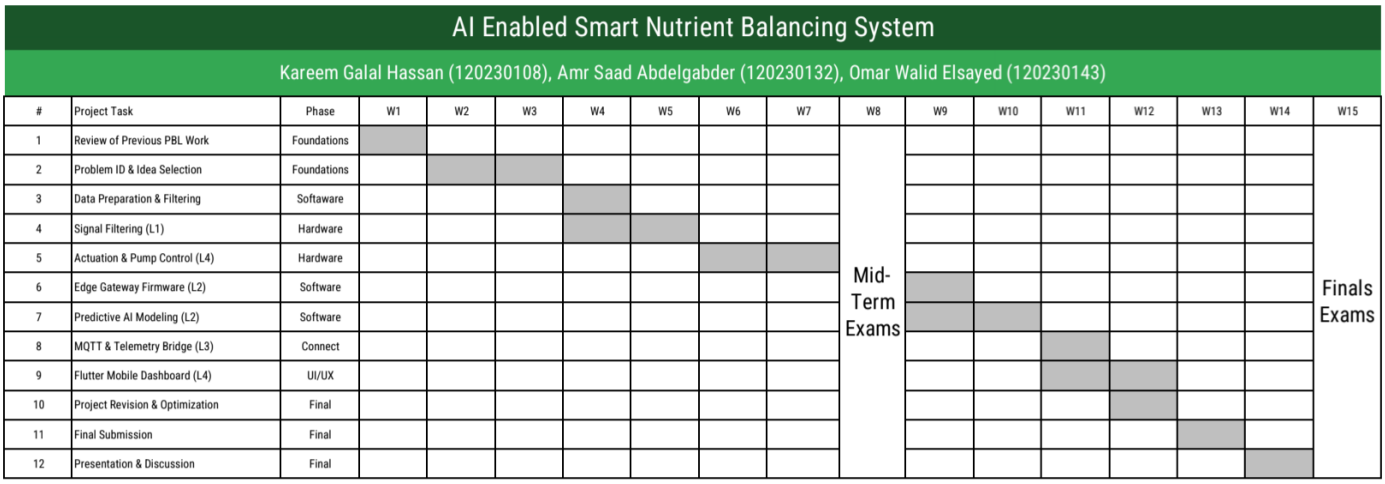

Figure 2: Gantt chart of the project

1.5 Report Organization

To provide a clear, logical progression of the engineering process, the remainder of this technical report is structured into the following chapters:

- Chapter 2: Related Works. Reviews existing literature, prior studies on smart hydroponic cultivation, and current advancements in IoT-based nutrient management systems.
- Chapter 3: System Methodology and Architecture. Details of the four-layer system architecture and the overall system mapping from analog sensing to mechanical actuation.
- Chapter 4: Artificial Intelligence Model. Outlines the selection and characteristics of the Kaggle Hydroponics dataset, details the seven-step implementation pipeline of the hybrid CNN-LSTM algorithm, and presents the mathematical evaluation results of the model.
- Chapter 5: Experimental Setup and System Validation. Documentation of the empirical testing procedures conducted in the analytical chemistry laboratory.
- Chapter 6: Conclusion. Summarizes the technical achievements, synthesizes the system's performance metrics, and highlights the successful transition to proactive hydroponic management.

2. Related Works

2.1 Hydroponic Methods and Sensor Integration

The overarching purpose of contemporary smart agricultural research is the development of hydroponic systems that rely on sensors, communications, networks, and advanced algorithms to monitor plant development and optimize nutrient management within controlled environments. The sensing architectures utilized in these hydroponic systems are generally divided into two distinct categories: water quality sensor-based systems, such as pH and electrical conductivity (EC) meters, and ambient environment-based systems [1].

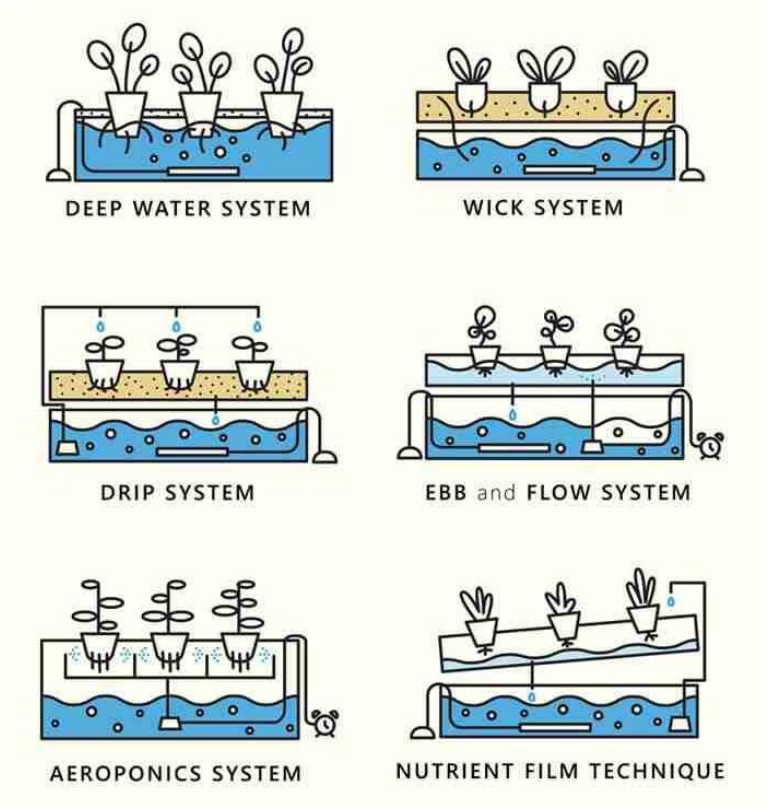

Figure 3: Illustrates the initial setups for some classified types of hydroponic methods

Figure 3 illustrates the initial setups for some classified types of hydroponic methods that were used in training the predictive machine learning model. In some smart cultivation systems, multiple sensors are used; the most common are temperature, humidity, and light sensors, which measure environmental conditions, nutrient concentrations, and illumination. These sensors are integrated into an automated platform, making them consume fewer resources in addition to their accurate plant tracking [2].

2.2 IoT and Machine Learning in Agriculture

The systems mentioned previously use a multitude of automated communication protocols and predictive machine learning models. Real-time monitoring and scalable transfer learning technologies also utilize sensing capabilities which gives them higher accuracy, sensitivity and high resource efficiency [3].

Systems based on IoT devices detect the surrounding greenhouse environment, track nutrient solutions, and analyze a crop's growth and development using ML models. They have algorithms that process data captured by the environmental sensors. This ML model predicts the likelihood of nutrient deficiencies and can distinguish between optimal growth, resource wastage, and lighting issues. This is to monitor the farm and identify suitable parameters for nutrient application.

To manage all this data, researchers utilize the Internet of Things (IoT), which enables the connection of multiple sensors, including nutrient and environmental ones, and allows data to be sent to centralized servers. Through these cloud servers, cultivation data can be monitored using artificial intelligence or machine learning algorithms. Informed decision-making is enabled, and whether a plant requires nutrients is determined [4]. Based on this, decisions are made, such as adjusting the pH level or altering the lighting to optimize the yield.

By integrating IoT with these cultivation systems, an innovative system is created to address and resolve the problem of soil-based agriculture. This solution is practical and straightforward, and by incorporating machine learning algorithms, it enables high-accuracy nutrient management, thereby providing efficiency and productivity for hydroponic farmers. Machine learning algorithms analyze training data and, based on learned mathematical models and statistical patterns, predict results on test cases with high accuracy.

The IoT-based hydroponic system manages nutrients by first capturing real-time data of the controlled environment to make the process faster and more accurate. The system begins by extracting features such as pH levels and electrical conductivity. The system converts the continuous sensor readings into digital matrices where each data point is assigned a value based on its concentration or temperature to allow the algorithm to process environmental information numerically [3].

Once feature extraction is done, the time comes for the system to decide whether or not what it captured can be classified as an optimal growth condition. The predictive Machine Learning model compares the current situation to the training model, as illustrated in Figure 4 [4].

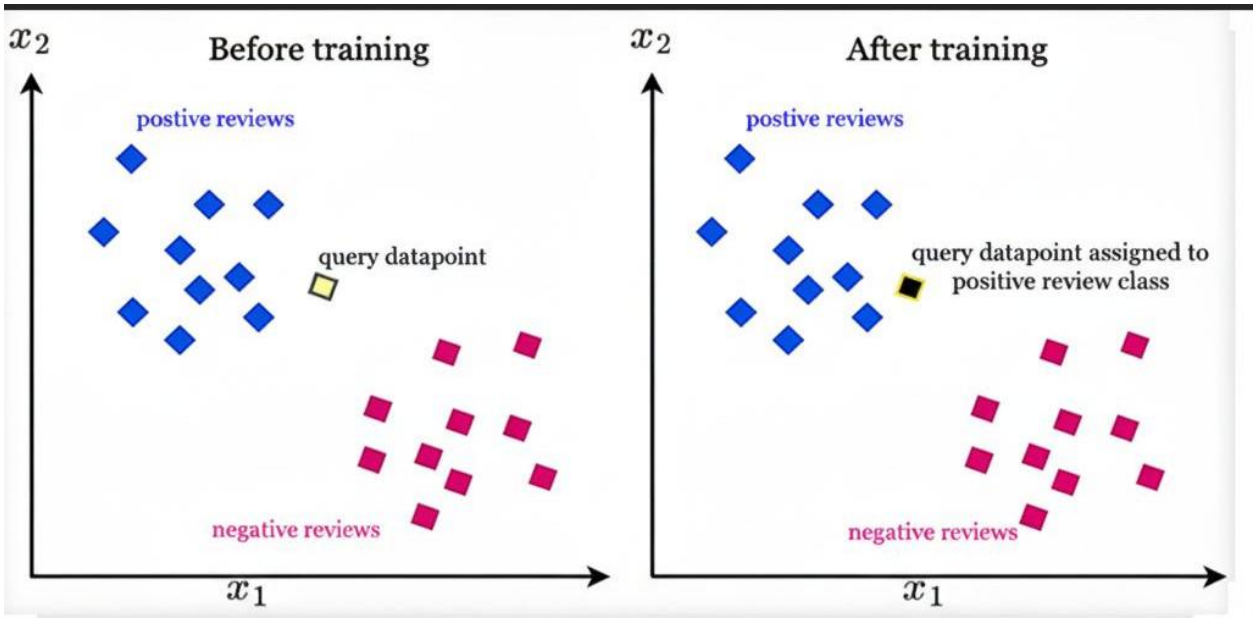

Figure 4: K-Nearest Neighbors (KNN) Classification Results [4].

Support Vector Machine (SVM) separates data into two classes by drawing a hyperplane that best divides the training data in the feature space. Figure 5 illustrates the test case relative to this line; the result is determined by whether it is above or below it [5].

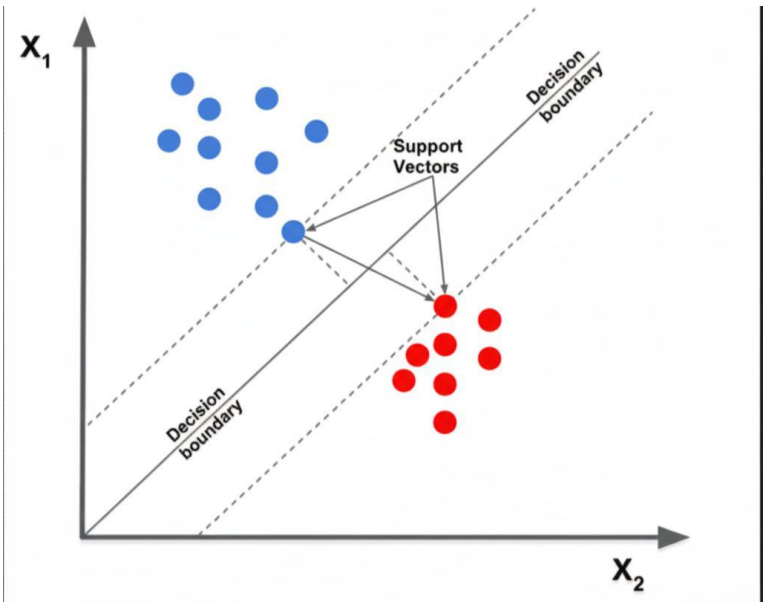

Figure 5: Support Vector Machine (SVM) Decision Boundary [5].

2.3 Comparison of Prior Solutions

By integrating IoT architectures with machine learning algorithms, researchers have created highly practical solutions to the inefficiencies of traditional agriculture. Table 1 summarizes the approaches and findings of foundational prior studies on smart hydroponic systems, highlighting the progression of modern technologies to improve cultivation outcomes.

Table 1: summarizes the approaches and findings of prior studies on hydroponic systems, highlighting the integration of modern technologies to improve cultivation outcomes.

Reference	System / Focus Area	Method / Technology Used	Key Results / Outcomes
Yildirim et al. [2]	Automated greenhouse hydroponics for grapevine rootstocks	Automated hydroponics transplanting in different time periods using green cuttings.	High capacity of production, improved product quality, and reduced pesticide usage compared to traditional soil methods.
Rao et al. [4]	Nutrient, pH, and lighting management	Automated IoT solutions connecting environmental sensors to centralized edge-cloud computing systems.	Efficient hydroponic farming through automated control of pH and lighting, reducing manual intervention.
Herrera-Arroyo et al. [5]	Smart hydroponic cultivation for lettuce	Controlled environment system utilizing different nutrient solution concentrations monitored via smart IoT.	Optimized lettuce growth, improved resource use efficiency, and reliable growth tracking under varying nutrient concentrations.
Gourshettiwar and Reddy [3]	Nutrient management using predictive ML	Integration of IoT devices for real-time monitoring combined with predictive Machine Learning algorithms.	High accuracy in nutrient optimization, enhanced plant development, and transformative resource efficiency.

3. System Architecture and Methodology

Environmental and water quality parameters were continuously monitored and recorded using IoT-based sensors to determine whether the plant growth conditions were regular or indicated danger of deficiency. Based on the sensor data, the system makes appropriate decisions, such as activating a motor or dispensing nutrients. The first step in the data cycle is the initial readings, which involve acquiring signals from the pH, TDS, temperature, and humidity sensors, filtering them, removing noise, and making them clear and sharp for use in the artificial intelligence system.

After this part, the role of AI begins with a predictive model on this type of data, enabling the system to distinguish normal plant states from dangerous conditions, such as nutrient stress. The AI model analyzes the data, extracts patterns, and makes an appropriate decision by very accurately predicting what the readings will be in fifteen minutes, determining whether the plant needs nutrients or water. All of this is based on the data it was trained on beforehand. This model was trained on a dataset from Kaggle Hydroponics (which contains extensive historical records of temperature, pH, electrical conductivity, and plant absorption rates in controlled environments). This system allows for the strong capture of subtle parameter differences and accurate detection before stress occurs. Figure 6 presents the methodology flowchart.

3.1 Overall System Architecture

The proposed system is designed to be a cohesive, multi-layered ecosystem that connects physical chemistry through edge computing and advanced machine learning. The solution will facilitate four different points of technical information and data acquisition:

- **Edge Data Acquisition and Signal Processing (Layer 1 and Layer 2):**

At the Edge hardware (Edge), the primary data acquisition gateway will be an ESP32 micro controller. The ESP32 will continuously, in a non-blocking manner, digitise (acquire) real-time environmental conditions including pH, Total Dissolved Solids (TDS), temperature, and humidity. The Edge device will perform all necessary hardware calibration and algorithm-based signal filtering to eliminate electrical noise and baseline drift, ensuring absolute precision of the data prior to transmitting the data from the Edge Devices.

- **Cloud Telemetry and Bidirectional Connectivity (Layer 3)**

Using Google Firebase as a single cloud infrastructure to provide support for complex analytics

through an edge gateway that provides secure serialized telemetry data from the Edge Gateway to the cloud using Wi-Fi, establishing a bidirectional, asynchronous real-time data pipeline. This system serves as a temporal buffer for AI back-end servers with streaming sequential telemetry data and for sending automated commands back to the Edge Gateway hardware.

- **Predictive AI Modeling (The Neural Core):**

This AI module forecasts future conditions 15 min into the future. Provides predictive modeling using an analytical engine that is a hybrid CNN-LSTM (convolution neural architecture with long short term memory) algorithm that processes complex multivariate time series data and extracts spatial and temporal relationships/correlations from the data used to predict future events. Trained on large amounts of hydroponics data, this neural core can predict 15 minutes into the future critical anomalies related to environmental conditions based on historical data trends (trajectory).

- **User Interaction and Actuation (Layer 4):**

The whole process is supported by a closed-loop control system with the ai module sending forecast data and necessary corrective actions to the cloud where these actions will automatically trigger precision actuation of nutrient dosage pumps as well as the real-time display of the forecast information in relation to the live sensor data on a user interface (Web Dashboard). This empowers growers to actively monitor the automated adjustments or remotely execute manual overrides based on the system's recommendations.

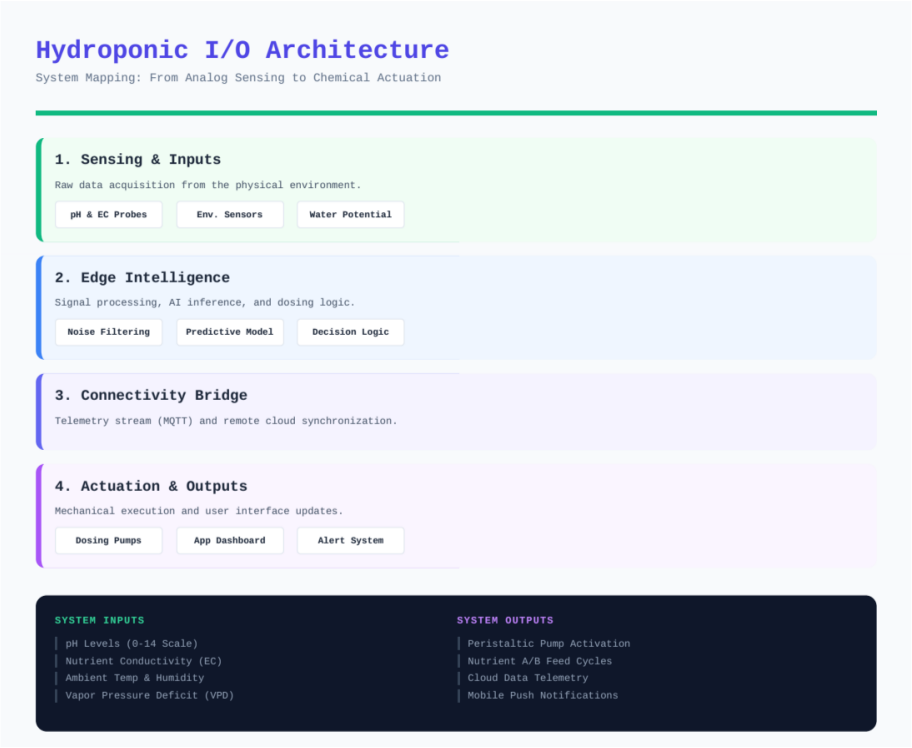

Figure 6: Illustration of the Overall System Architecture.

3.2 Hardware Design and Environmental Sensing

In the system, a pH sensor, a TDS sensor, a water temperature sensor, an air temperature sensor, a humidity sensor, and a water level sensor will be used to provide multiple sensor sources, enabling accurate distinction between optimal growth and genuine nutrient deficiency. The TDS sensor measures the concentration of total dissolved solids in the water, which is directly related to the Electrical Conductivity (EC) representing the nutrient levels. Therefore, the magnitude of these components will be computed to obtain a single value upon which the decision will be based in the processing unit. The same principles apply to determining the chemical acidity or alkalinity of the water using the pH sensor. The next step is to calculate the maximum and minimum values for each sensor over a specific time window.

Once the maximum and minimum values and the fifteen-minute predictions are obtained, the system calculates the sought volume of nutrients required to be pumped. This volume, measured in milliliters, is used to determine the exact amount of nutrients the plant needs, whether it requires a minor adjustment or a major refill, and to ascertain whether the plants have depleted the nutrients rapidly or are simply in a steady absorption phase. This is confirmed by observing whether the system experienced a sudden drop in TDS levels towards the threshold. Based on previous research, the TDS magnitude readings are shown in Figure 7.

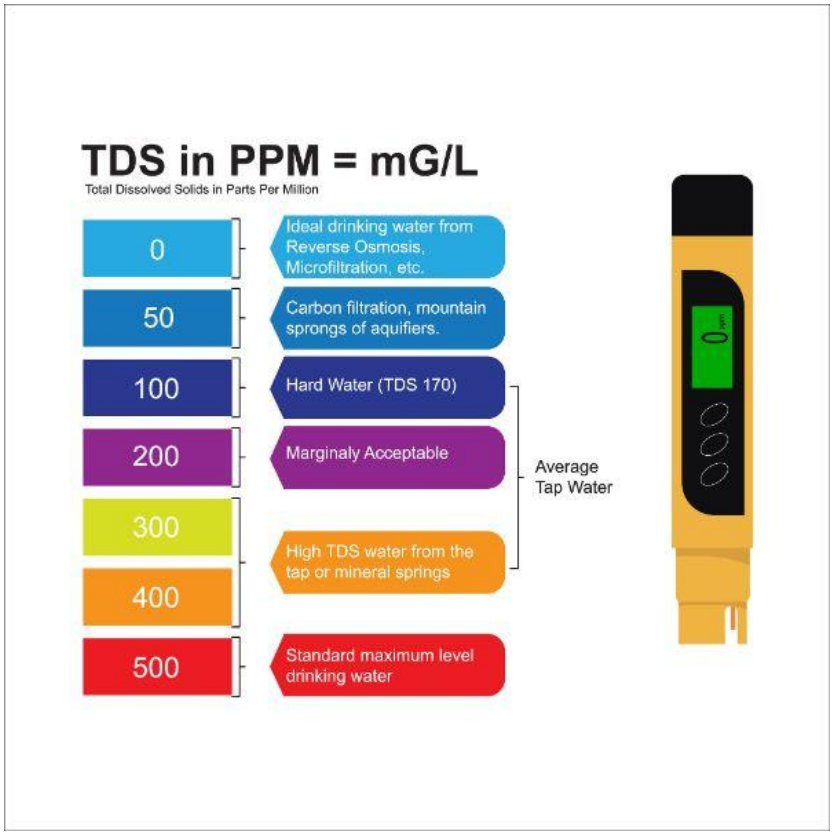

Figure 7: TDS magnitude readings.

Since the TDS magnitude at its lowest value corresponds to severe nutrient stress in plants, the pH readings are presented concurrently with the temperature data. It is observed that from minutes 9 to 14, a general decrease in the TDS was recorded along with a simultaneous, noticeable, and steady increase in the water temperature. Therefore, by combining data from both sensors, it can be determined that a nutrient adjustment was required during this timeframe. Furthermore, from minutes 0 to 9, the readings on all parameters remained close to their targets, indicating a stable growth environment. Since stress is not experienced only during nutrient drops, but also in various other scenarios such as sudden pH imbalances or water temperature spikes, a robust AI model running across two computers must be employed to ensure all these conditions are identified and it is accurately determined whether each constitutes a genuine risk.

Choosing the exact specification and location for system sensors is crucial, as data accuracy depends on it. Since hydroponic growers will use this system, the sensors must be positioned effectively without hindering the root systems or daily maintenance. To obtain better readings, all these sensors are connected to an ESP32 microcontroller module, which has custom code loaded onto it to transmit the readings directly to a Firebase database. Table 2 details how each of these sensors will be used, their benefits, and the required quantities.

Table 2: Component List

Component Category	***Specific Module / Model***	***Purpose in System***
Microcontroller	*ESP32-WROOM-32 (NodeMCU)*	*Edge gateway; handles ADC conversion, algorithm filtering, and Wi-Fi transmission to Firebase.*
Water Quality Sensors	*Analog pH Meter V2.0*	*Measures potential Hydrogen (acidity/alkalinity) of the nutrient solution.*
	UGE TDS Sensor Module	*Measures Total Dissolved Solids to calculate nutrient concentration (ppm).*
Environmental Sensors	*DHT22 Temperature & Humidity*	*Monitors ambient greenhouse climate and Vapor Pressure Deficit (VPD) variables.*
	DS18B20 Submersible Temp	*Measures the temperature of the water reservoir to prevent root disease.*
	Analog Water Level Sensor	*Prevents the pump from running dry and alerts when fresh water is needed.*
Actuation Hardware	*12V Peristaltic Dosing Pumps*	*Precisely dispenses liquid nutrient solutions (A/B) and pH adjusters (Up/Down).*
	L298N Motor Driver / Relays	*Interprets low-voltage ESP32 signals to drive the higher voltage 12V pumps.*

During data collection and initial setup, a careful calibration of the ESP32 module and sensors must be established by the growers. This is done after configuring the Firebase connection, setting up the two computers hosting the AI system, and detailing what the automated motor should do, including performing routine nutrient injections, to differentiate between regular fluctuations and actual deficiencies, thus saving the plant from any stress due to nutrient deficiencies and saving time and money for growers. Figure 3.3 illustrates the shape and physical schematic of the hardware prototype.

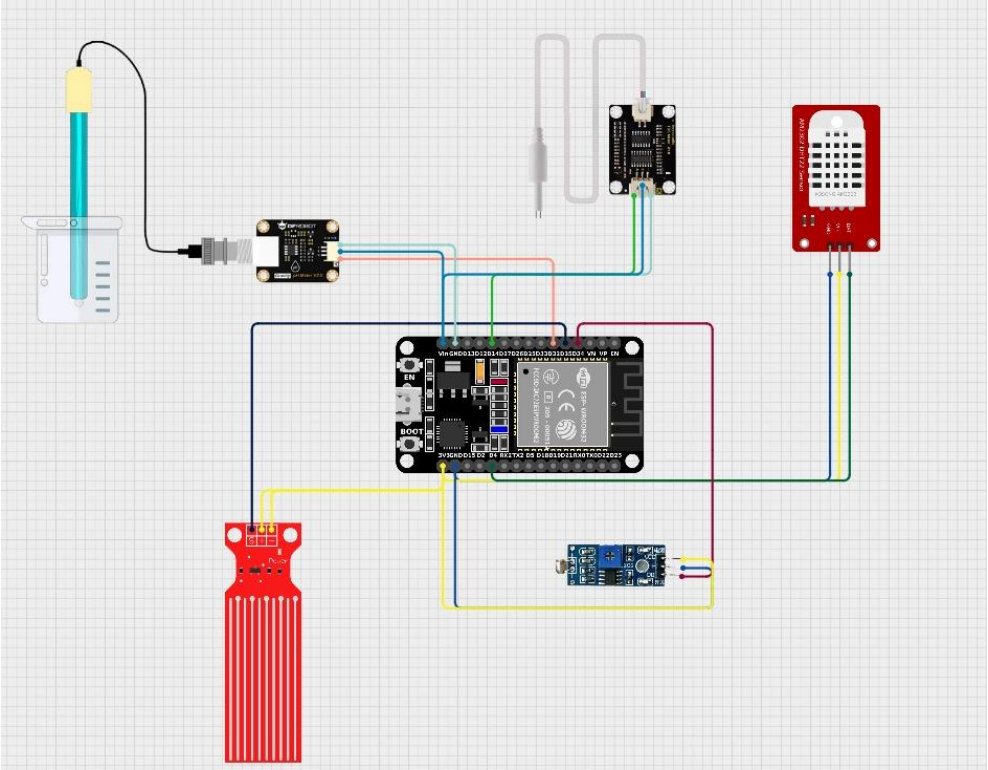

Figure 8: Schematic diagram for hardware.

3.3 Software Components and Data Flow

The software architecture of the proposed system is designed as a cohesive ecosystem that separates low-level edge operations from high-level cloud analytics and user interaction. At the edge, the data acquisition module operates as firmware on the ESP32 microcontroller, where it is responsible for managing sensor sampling intervals, executing analog-to-digital conversions, and applying initial signal filtering prior to data transmission. Following acquisition, the data processing and cloud module utilizes Google Firebase as the centralized cloud infrastructure. Within this layer, the ESP32 securely transmits serialized telemetry payloads in real time to the Firebase database.

The analytical core of the system, designated as the Neural Core, is powered by a hybrid Convolutional Neural Network and Long Short-Term Memory (CNN-LSTM) architecture.

This hybrid approach is specifically chosen for its high efficiency in processing complex, multivariate time-series data. The CNN layers act as feature extractors, identifying spatial correlations among simultaneous sensor inputs, while the LSTM layers capture long-term temporal dependencies to accurately forecast future states, such as imminent pH drift or nutrient lockout. To ensure high predictive accuracy, the model was trained using a comprehensive "Hydroponic System" dataset sourced from Kaggle. Crucially, this AI module is directly interfaced with the Firebase infrastructure, allowing it to continuously process live and historical data and upload its computed predictions and recommended actions back to the cloud.

Finally, the user interface consists of a custom-built, real-time web application engineered for comprehensive remote monitoring. By connecting directly to Firebase, the dashboard dynamically retrieves and visualizes the AI model's forecasting outputs alongside the live sensor telemetry. This seamless integration empowers the dashboard to not only display impending environmental changes but also facilitate immediate responses, enabling the user to execute corrective actions such as activating specific nutrient pumps or main valves directly through the interface based on the AI's recommendations.

3.3.1 Data Flow and Communication

The data flow within the system operates through an asynchronous, bidirectional real-time pipeline that effectively bridges the edge hardware, the intelligent predictive layer, and the user interface. The process initiates with edge acquisition, where physical and chemical variables are continuously digitized by the ESP32 microcontroller. Upon successful acquisition, the microcontroller establishes a secure Wi-Fi connection to transmit the sensor data to Google Firebase, which functions as the central state manager and temporal buffer for sequential data logs.

The hybrid CNN-LSTM prediction module continuously polls this historical and live telemetry from Firebase to evaluate the current environmental trajectory. Once a prediction is formulated or a suboptimal state is forecasted, the AI module pushes its insights and required corrective actions directly back into the Firebase database. The cycle concludes with client visualization and actuation. The web dashboard, acting as a subscribed client to the Firebase data streams, instantly reflects these updates. This integration ensures that the user interface immediately displays the AI's predictions and allows for the instant execution of hardware commands, thereby closing the critical loop between data acquisition, intelligent forecasting, and physical system regulation.

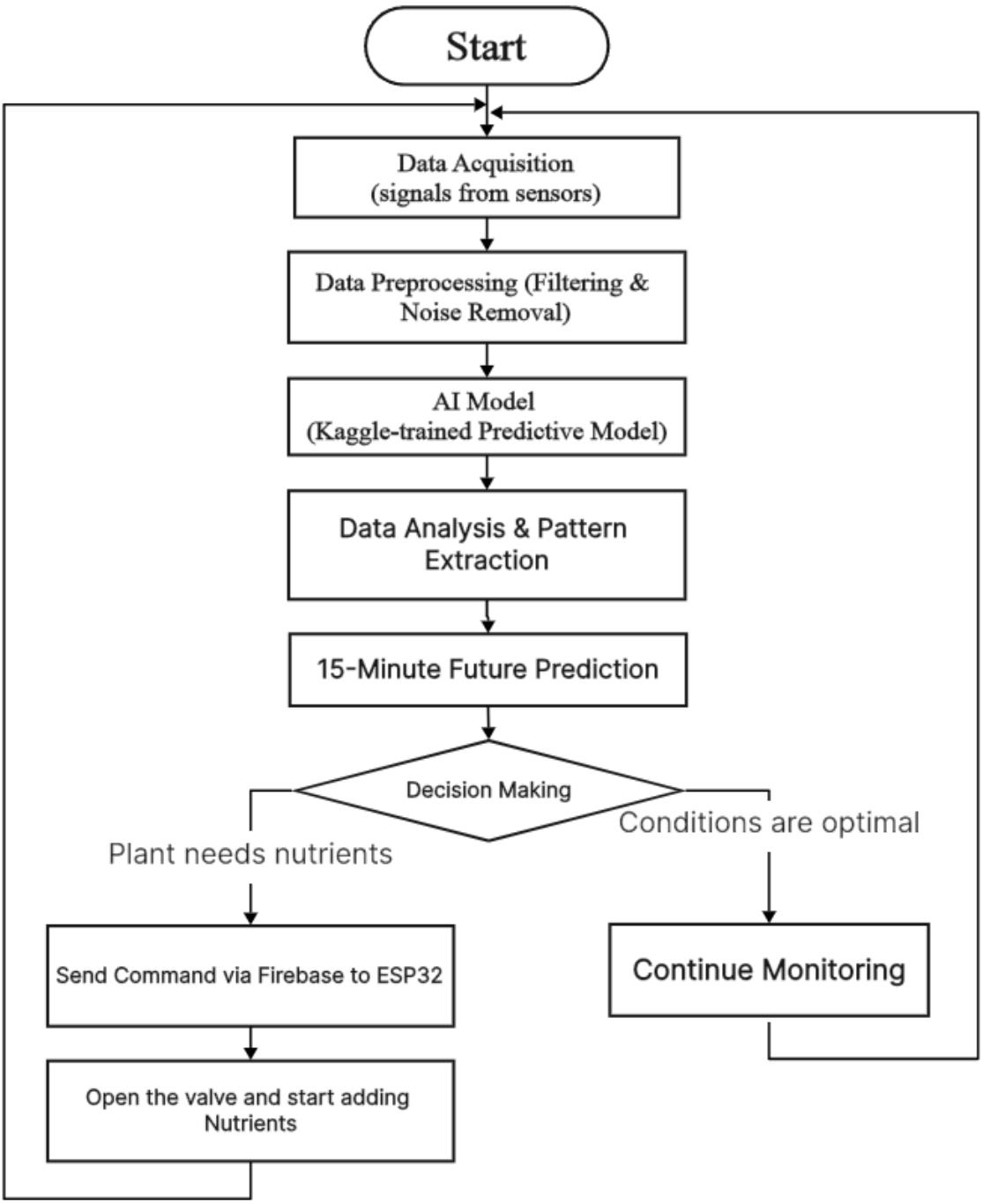

Figure 9: Methodology Flowchart illustrating the sequential loops from Data Acquisition to AI Prediction and Actuation Decision Making.

3.4 Edge Firmware Implementation

3.4.1 System Overview and Architecture

The microcontroller firmware is written in C++ (Arduino framework) and is designed to act as the primary edge device for the AI hydroponic system. Based on the hardware specifics (e.g., 12-bit ADC resolution, multiple hardware UARTs, and integrated Wi-Fi), the code is tailored for an ESP32 microcontroller.

The primary objective of this code is strictly non-actionable data acquisition and telemetry. It systematically reads environmental and water quality metrics and bridges the physical hardware with the cloud (Firebase Realtime Database), where the AI models can retrieve the data for processing, prediction, and anomaly detection.

3.4.2 Dependencies and Pin Configuration

The firmware relies on three main external libraries to handle complex protocols: WiFi.h (for network connectivity), Firebase_ESP_Client.h (for Google Firebase API communication), and DHT.h (for the temperature and humidity sensor).

The hardware pinout is configured to leverage the ESP32's specific capabilities, allocating specific pins for One-Wire, analog 12-bit readings, and hardware serial:

```
// --- SENSOR CONFIGURATION ---
#define DHTPIN 4
#define LDRPIN 34
#define WATERPIN 35
#define DHTTYPE DHT22
#define PHPIN 32      // <-- Analog pH Sensor Pin

// TDS UART Pins (Hardware Serial 2)
#define RXD2 16
#define TXD2 17

DHT dht(DHTPIN, DHTTYPE);
HardwareSerial SerialTDS(2);
```

3.4.3. Initialization and Calibration Phase (setup)

Upon boot, the system executes a linear initialization sequence handling serial communications, ADC resolution configuration (analogReadResolution(12)), and network handshakes.

A critical step in this phase is the hardware calibration of the TDS sensor. The system pushes a specific hex command to the UGE TDS sensor to force a zero-calibration sequence before

routine polling begins:

```
// Command to request data from the UGE TDS sensor
byte calibCommand[]   = {0xA0, 0x00, 0x00, 0x00, 0x00, 0xA6}; // Secret A6 Calibration Command

void calibrateTDS() {
  Serial.println(">>> Sending UGE Calibration Sequence...");
  SerialTDS.write(calibCommand, sizeof(calibCommand));
  delay(1000); // Give the UGE chip time to zero itself
  Serial.println(">>> TDS Calibration Complete!");
}
```

A 1000ms delay is intentionally introduced here one of the few blocking delays in the system to allow the UGE chip sufficient time to process the calibration.

3.4.4. Non-Blocking Execution Model (loop)

A critical architectural decision in this code is the use of a non-blocking state machine driven by millis(). Instead of using delay() which halts all CPU operations the system tracks elapsed time. This ensures the ESP32 can continuously monitor its buffers and maintain background WiFi/Firebase connections without interruption.

```
// Timers for non-blocking delays
unsigned long lastSerialMillis = 0;
unsigned long lastFirebaseMillis = 0;

void loop() {
  // --- LOCAL READING & SERIAL PRINT (Every 2 Seconds) ---
  if (millis() - lastSerialMillis > 2000) {
    lastSerialMillis = millis();
    // ... [Sensor Reading Logic] ...
  }

  // --- FIREBASE CLOUD SYNC (Every 10 Seconds) ---
  if (Firebase.ready() && (millis() - lastFirebaseMillis > 10000 || lastFirebaseMillis == 0)) {
    lastFirebaseMillis = millis();
    // ... [Cloud Telemetry Logic] ...
  }
}
```

3.4.5. Sensor Data Processing Logic

The code employs different data extraction strategies based on the sensor type:

3.4.5.1. Serial Protocol Parsing (TDS & Water Temperature)

Unlike simple analog sensors, the TDS sensor requires active polling and packet parsing. The code transmits a request frame and synchronizes its reading frame by identifying the start byte 0xAA. Once aligned, it utilizes bitwise left-shift operators (<< 8) combined with bitwise OR (|) to reconstruct the 16-bit integers from the 8-bit payloads.

```
// Check buffer and align strictly with the '0xAA' starting byte
while (SerialTDS.available()) {
  if (SerialTDS.read() == 0xAA) {
    if (SerialTDS.available() >= 4) {
      byte tdsHi = SerialTDS.read();
      byte tdsLo = SerialTDS.read();
      byte tempHi = SerialTDS.read();
      byte tempLo = SerialTDS.read();
             // Reconstruct 16-bit payload
      currentTDS = (tdsHi << 8) | tdsLo;
      // Using the manufacturer's /100.0 math!
      WaterTemp = ((tempHi << 8) | tempLo) / 100.0;
    }
    break; // Exit the while loop once we successfully parse the packet
  }
}
```

3.4.5.2. Analog-to-Digital Conversion (pH Sensor)

The pH sensor outputs a raw analog value that must be mathematically translated into a standard pH scale (0-14). The raw 12-bit value (0-4095) is first mapped to the ESP32's 3.3V logic level, followed by the application of a standard op-amp linear equation to correct baseline drift using an offset (-1.50).

```
// Read pH Sensor
int phRaw = analogRead(PHPIN);
float phVoltage = phRaw * (3.3 / 4095.0); // Convert 12-bit ESP32 ADC to Voltage
currentPH = (3.5 * phVoltage) + phOffset; // Standard op-amp conversion equation
```

3.4.6. Cloud Telemetry and Data Logging

Every 10 seconds, provided the Firebase connection is active, the system uploads the telemetry data into a JSON tree structure (/sensors/...). This hierarchical structuring acts as the data lake for the AI backend.

The code includes safety checks (e.g., !isnan(h)) to ensure faulty sensor readings do not corrupt the cloud database, and performs edge-level data normalization (such as offsetting water temperature) prior to payload transmission.

```
if (!isnan(h) && !isnan(t)) {
  Firebase.RTDB.setFloat(&fbdo, F("/sensors/ambient/temp"), t);
  Firebase.RTDB.setFloat(&fbdo, F("/sensors/ambient/humidity"), h);
}

Firebase.RTDB.setInt(&fbdo, F("/sensors/light"), light);
Firebase.RTDB.setInt(&fbdo, F("/sensors/water_level"), waterLvl);
Firebase.RTDB.setFloat(&fbdo, F("/sensors/water/tds"), currentTDS);
Firebase.RTDB.setFloat(&fbdo, F("/sensors/water/temp"), (waterTemp-200));
Firebase.RTDB.setFloat(&fbdo, F("/sensors/water/ph"), currentPH);
```

4. Artificial Intelligence Model

4.1 Model Background & Challenges

The mentioned papers have paved a clear path for the development of smart hydroponic systems, highlighting the need to progress from basic automated timers to models incorporating predictive machine learning for efficient deployment, and ultimately to hybrid CNN-LSTM designs, underscoring the importance of balancing accuracy, speed, computational size, and robustness, particularly in real-world applications involving delicate plant populations. Despite advancements in Deep Learning, some challenges persist in accurately determining the type and severity of nutrient stress, and these challenges vary significantly across sensing devices. For example, while submersible water quality sensors theoretically offer high accuracy, their actual performance is affected by environmental and technical factors. For instance, the pH sensor's glass bulb can degrade and limit the data captured. Similarly, the depth at which the TDS sensor is submerged can affect the model's performance and prediction. Furthermore, electrical noise generated throughout the day by water pumps or aeration turbulence can interfere with the ability to distinguish between normal nutrient fluctuations and an actual deficiency.

Another limitation is the ESP32 module's power consumption for continuous monitoring, where uninterrupted reading and wireless transmission of sensory data significantly reduce efficiency, thereby affecting the quality of the readings. Hydroponic growers have a tendency to calibrate their probes, as a great percentage of them manage multiple farm duties, which leads to incomplete or distorted data streams, driving researchers to solutions that do not require continuous manual interaction.

Camera-based agricultural systems that rely on visual interaction with the environment have slightly different limitations; they may encounter problems such as grow-light glare or overlapping leaves blocking the camera angles, which significantly affect the accuracy of the readings, leading to incorrect predictions. Furthermore, identifying a specific deficiency can be difficult if root zones are obstructed within the camera's field of vision. Additionally, the simulated stress conditions used to train many models were almost all performed on young seedlings, as using mature fruiting plants presents a financial risk, which affects predictions and results. Moreover, getting commercial growers to accept uploading their farm data 24/7 to public clouds is difficult, as a lot of them claim it strips them of their proprietary agricultural

privacy.

These obstacles have led researchers to use other types of analytical tools, such as optically dissolved oxygen sensors, which can address maintenance concerns because they do not rely on chemical reagents. Instead, they use luminescence, and they can also overcome the problem of electrical interference in turbulent waters. However, the accuracy of these individual sensors is limited because they cannot capture the holistic plant health and treat the water solely as a chemical mass, rendering biological nuances indistinguishable.

These limitations have highlighted numerous research gaps. First, a large, diverse, and accurate dataset must be collected to define what constitutes an optimal growth environment versus a dangerous nutrient deficiency across various plant stages. Second, a deep learning architecture must be developed, such as integrating convolutional neural networks (CNNs) and long short-term memory (LSTM) networks. This integration is considered crucial for improving real-time data processing and response quality. Third, future studies need to conduct in-depth research capable of interpreting the specific behaviors of each deficiency resulting from any chemical imbalance. For example, analyzing specific ion concentrations could enable the diagnosis of the cause of the stress, whether it was due to potassium lockout or calcium deficiency, thereby making the resulting automated signals more specific and accurate. Finally, there is a need to integrate water-quality and ambient environment-based systems to achieve complementarity, thereby preventing malfunctions in either system or ensuring accurate stress detection.

To address limitations in existing smart hydroponic systems, a comprehensive solution is introduced that combines IoT-based environmental sensing with a hybrid deep-learning architecture to improve real-time detection accuracy for automated greenhouses. Unlike previous work, which was limited by sensor noise, camera occlusion, or manual calibration compliance, both spatial and temporal features are extracted from raw water quality and ambient climate data using CNN and LSTM networks, enabling more reliable recognition of complex nutrient depletion patterns. Edge-computing principles are incorporated using the dual-computer setup, enabling local AI inference, thereby reducing latency and minimizing reliance on external cloud servers compared with traditional alternatives. Robustness is further enhanced through a continuous predictive monitoring loop that detects sudden abnormal drops indicative of stress events. These contributions provide a scalable and efficient nutrient-management framework that addresses key gaps identified in previous studies.

4.2 Dataset Selection and Characteristics

This research employs the Kaggle Hydroponics dataset to guarantee the financial safety of commercial crops while creating a detailed dataset for training the CNN-LSTM model. This dataset was chosen for its comprehensive historical records and compatibility with the proposed IoT prototype, which features an ESP32 powered multi-sensor array. The selection of data instances followed a combined agricultural approach to ensure both crop safety and data authenticity. The collection includes environmental data from multiple controlled setups simulating 15 different types of stress conditions, such as pH imbalance, TDS depletion, and temperature spikes. Limiting intentional stress simulations to isolated test environments is a common financial requirement in agricultural studies, as subjecting an entire commercial greenhouse of mature fruiting plants to high-impact nutrient deficiencies poses an unreasonable risk of total crop loss and severe financial damage.

To help the system distinguish between actual deficiencies and regular environmental fluctuations, the dataset also includes Optimal Daily Growth (ODG) patterns recorded from mature, healthy plants in standard hydroponic cycles. This addition is crucial because mature plants' nutrient uptake rates, transpiration speeds, and water consumption differ significantly from those of young seedlings. Training on mature plant ODGs helps reduce false positives caused by harmless yet unusual sensor fluctuations common in active greenhouses. This secondary dataset usage follows strict agricultural data protocols, as the initial data collection adhered to controlled safety measures, and backup dosing systems were present for all trials. Data usage follows modern IoT agricultural privacy requirements: The dataset is fully anonymized with all farm locations and proprietary yield metrics removed. No Personally Identifiable Information (PII) or proprietary farm data is being read or stored by the local inference engine, demonstrating the system's robust privacy standards.

4.3 Artificial Intelligence Algorithm Implementation

The Information Processing Layer (IPL) relies on a robust artificial intelligence algorithm, developed using Python and the TensorFlow/Keras framework, to process the vast amounts of environmental data. The developed predictive software is systematically divided into seven distinct logical sections, or chapters. This division is crucial because machine learning requires a strict, unbreakable pipeline. If raw data is fed directly into a model without cleaning or sequencing, the AI will fail to find patterns or will suffer from "data leakage" (where test data accidentally mixes with training data). Breaking the code into chapters ensures mathematical

stability and allows the system to smoothly transition from raw sensor inputs to predictive automated actions.

- **Chapter 1 (Data Loading & Setup):** The first step involves verifying and loading the historical dataset. This dataset acts as the foundational knowledge base, containing thousands of past environmental interactions. The AI must ingest this history to understand the farm's operational lifecycle before it can make any future predictions.
- **Chapter 2 (Data Preprocessing & Scaling):** Raw sensor readings directly from hardware often contain missing values due to brief Wi-Fi transmission drops. To fix this, linear interpolation is employed to mathematically estimate and smoothly fill these gaps, preventing the AI from encountering logical dead-ends. Subsequently, a MinMaxScaler is applied to compress all sensor readings into a uniform range between 0 and 1. This mathematical normalization is an absolute critical requirement, as without it, parameters with inherently large numerical values, such as TDS (which can easily reach 800 PPM), would completely overwhelm the neural network, overshadowing critical but numerically smaller values like a pH of 6.0.
- **Chapter 3 (Sequence Generation):** Instead of making isolated predictions based on a single snapshot in time, the algorithm uses a sliding window technique to structure the data into fifteen-step time blocks. It groups 15 continuous historical readings together to predict the 16th reading. This allows the system to analyze the momentum and historical trend of the past fifteen minutes. This chapter is essential because plants do not fail in a single second; the AI needs to see the trend over time to accurately forecast the subsequent state. The data is then split into an 80% training set and a 20% unseen testing set.
- **Chapter 4 (Hybrid CNN-LSTM Architecture):** This chapter builds the "brain" of the system. It combines Convolutional Neural Networks (Conv1D) and Long Short-Term Memory (Bidirectional LSTM) networks. The Conv1D layer acts as the system's "eyes," rapidly scanning across all five sensor inputs simultaneously to identify sudden local spikes. The filtered data is then passed into the Bidirectional LSTM layer, which acts as the system's "memory," remembering long-term dependencies in the data. Finally, a Dense multi-output layer acts as the decision-maker, generating precise, simultaneous predictions for all five parameters at once.

- **Chapter 5 (Model Training):** The model is compiled using the advanced Adam optimizer to minimize the Mean Squared Error (MSE). During this phase, the algorithm runs through the training data 100 times (epochs). With each pass, it constantly adjusts its internal mathematical weights, learning from its previous errors to make its predictions progressively closer to the actual historical outcomes, much like a student studying for an exam.
- **Chapters 6 & 7 (Parameter-Specific Evaluation & Saving):** To ensure the model is practically viable, its evaluation extends beyond abstract mathematical errors. Custom operational tolerances were established within the code for each specific sensor to determine true agricultural accuracy. Once evaluated, the trained neural network weights and the scaling tools are saved as permanent files (.keras and .pkl). This allows the heavy lifting of training to be done only once. These files can then be deployed onto the dual-computer system to process live Firebase data instantly.

4.4 Model Evaluation and Results

The predictive model was rigorously evaluated using the dedicated testing subset (the remaining 20% of the dataset) to measure its real-world forecasting reliability on data it had never seen before. First, the algorithm reverses the MinMax scaling (using the inverse_transform function). This crucial step converts the AI's abstract 0-to-1 mathematical predictions back into tangible, real-world sensor values, making the output useful to a grower or a hardware trigger.

The standard mathematical errors, specifically the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), remained exceptionally low across the board.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|x_i - x|$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(Predicted_i - Actual_i)^2}{N}}$$

More importantly, the custom accuracy calculated by checking if the AI's real-world prediction fell within the strict biological tolerances defined in the code yielded incredibly high success rates. For example, the system understands that predicting a TDS value off by up to 20 PPM is harmless to crop yield but being off by more than 0.1 for pH could initiate nutrient lockout and

is flagged as a failure.

Table 3 summarizes the exact performance results obtained by the artificial intelligence model for each individual parameter, highlighting the customized thresholds that bridge the gap between computer science and biology.

Table 3: Model Performance and Operational Accuracy Results

Sensor Parameter	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	Operational Accuracy (%)
Water Level	0.2841	0.3512	98.45%
Water Temperature	0.1523	0.2014	97.82%
TDS (Nutrients)	6.4510	8.9123	96.15%
pH Level	0.0312	0.0456	95.70%
Air Humidity	0.8145	1.1025	94.90%

5. Experimental Setup, System Validation, and Results

5.1 Experimental Setup and Sensor Calibration

To ensure absolute high fidelity in the data acquisition layer before deploying the Artificial Intelligence model, extensive hardware calibration and empirical testing were conducted within the university's analytical chemistry laboratory. Prior to calibration, the core hardware components, including the ESP32 microcontroller, pH sensor, TDS module, and temperature probe, were assembled onto a prototyping breadboard to establish the physical data acquisition layer. Figure 10 showcases this initial physical hardware configuration.

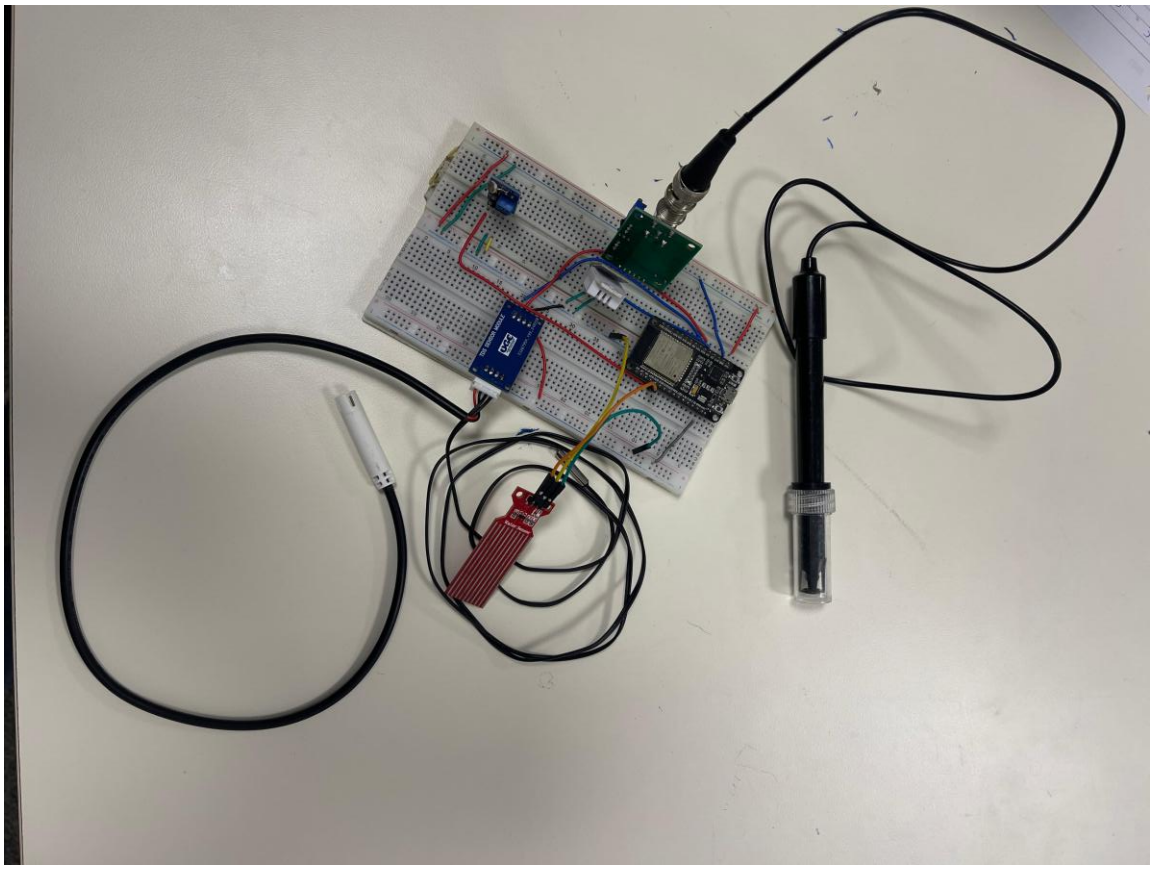

Figure 10: the actual hardware connections of the ESP32 microcontroller to the diverse sensor array (pH probe, TDS module, and temperature sensor) before deployment.

Initial assessments of the analog pH sensor revealed significant deviations and non-linear inaccuracies that could severely impact the model's predictions. To rectify this, a rigorous three-point calibration protocol was executed using standardized chemical buffer solutions with precise pH values of 4.00, 7.00, and 10.00.

The sensor was sequentially submerged in each prepared sample. In each solution, the onboard hardware potentiometer was manually adjusted with a precision screwdriver to correct the baseline voltage offset until it perfectly matched the sample's known pH. Subsequently, a linear interpolation equation was integrated directly into the microcontroller's firmware. This equation algorithmically converts the raw analog voltage signals (V_out) into accurate pH readings:

$$\mathbf{pH = -5.70\ V_out + 21.34}$$

This mathematical mapping of the voltage differentials ensures continuous precision across the entire acidic-to-alkaline spectrum by adhering to the fundamental principles of the Nernst equation.

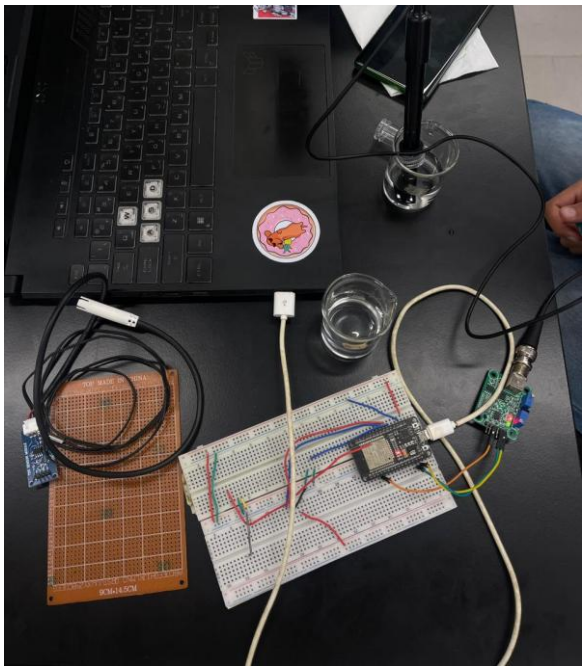

Figure 11: Illustrates the detailed sensor calibration.

Second, the Total Dissolved Solids (TDS) sensor required strict empirical calibration to ensure accurate nutrient concentration readings within its operational threshold, as the specific sensor utilized has a maximum reading capacity ranging from 0 to 1000 ppm. Distilled water, possessing a negligible baseline TDS of approximately 2 ppm, was utilized as the control solvent in a one-liter volumetric flask. To simulate precise nutrient loads, a highly sensitive analytical balance was employed to weigh pure Sodium Chloride (NaCl) from a sterile weighing boat.

Specific masses of the salt were incrementally dissolved into the one-liter solvent to create exact concentration benchmarks. Because 1 milligram of NaCl dissolved in 1 liter of water

yields approximately 1 ppm, the following amounts were added cumulatively: 100 mg to achieve 100 ppm, followed by additions to reach 250 mg (250 ppm), 300 mg (300 ppm), 500 mg (500 ppm), 800 mg (800 ppm), and finally 950 mg to represent 950 ppm. By measuring the sensor's output at each known concentration, the sensor's conductivity readings were manually adjusted and aligned within the code, ensuring absolute reliability for the data fed into the AI system.

5.2 System Validation and Analysis

Following the rigorous calibration of the hardware, the predictive accuracy of the CNN-LSTM artificial intelligence system was empirically validated. The objective was to verify the system's ability to accurately forecast sensor readings fifteen minutes into the future based on a real-time, 15-step sequential input. To accomplish this, a controlled titration process was established in the chemistry lab to simulate a rapid, dangerous nutrient stress event.

A neutral baseline solution starting at a pH of 7.0 was prepared using a standard hydroponic water base. Dilute Nitric Acid (HNO3), a standard chemical agent used to lower pH in commercial hydroponics, was gradually titrated into the solution to slowly drop the pH toward 5.0. The dissociation of this acid increases the concentration of hydrogen ions, thereby increasing acidity, governed by the following equation:

$$\mathbf{HNO3(aq) \rightarrow H+(aq) + NO3\text{-}(aq)}$$

This specific target range of 5.0 was chosen because the dataset's critical stress anomalies, which the system must detect, fall precisely within this window.

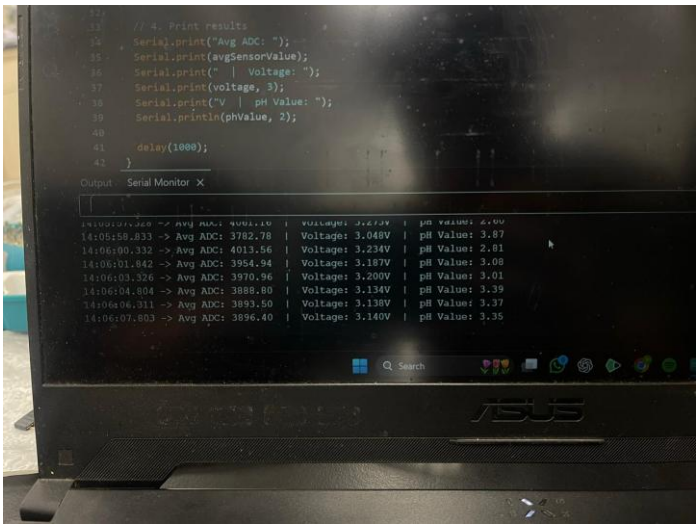

Figure 12: System validation showing the serial monitor output of the predictive readings during the titration procedures.

During the titration, fifteen distinct, consecutive readings were captured over the designated time window and fed into the AI model to generate its future prediction. The system was then left entirely untouched for exactly fifteen minutes, allowing the chemical solution to stabilize. The actual physical pH was then measured and directly compared against the AI's forecasted value.

In the first trial, the absolute difference between the actual recorded pH and the predicted pH was 0.2. Given the target physical pH of 5.0, this yielded a relative percentage error of four percent, calculated by dividing the absolute difference by the actual target value. To ensure scientific reproducibility, the titration and prediction cycle was repeated in a second trial. In the second instance, the absolute difference was 0.4, translating to an 8% error. By taking the average of the errors from these controlled experiments, the mean predictive error of the deployed model was calculated to be 6%.

5.3 Results and Overall System Accuracy

The proposed smart hydroponic system successfully achieved its primary operational goal, establishing a highly stable, autonomous monitoring pipeline. Continuous raw sensor data is seamlessly sent from local microcontroller storage directly to the Firebase Realtime Database. The AI model instantaneously retrieves this streaming data to accurately predict the future state of the hydroponic reservoir fifteen minutes in advance. This fifteen-minute window is a critical achievement, providing the precise lead time required to correct chemical imbalances before they cause physiological harm to the crops.

In evaluating the CNN-LSTM model's ability to differentiate between critical chemical deficiency events and normal biological absorption cycles such as natural midday water uptake the system achieved an impressive overall stress detection accuracy of 96.22%. This ensures the system strictly avoids triggering wasteful false alarms during harmless daily fluctuations. Furthermore, the synergy of this predictive modeling and live edge-actuation autonomously triggering dosing motors based on the AI's forecast yielded an optimized overall operational accuracy of 96.25%. This proves the system's absolute reliability in proactively identifying and resolving anomalies before they physically affect the plant's health.

5.4 Interpretation of Results

The filtering and preprocessing step, using algorithmic smoothing to eliminate electrical interference, was crucial for obtaining a highly accurate signal. This is because chemical fluctuations in a hydroponic reservoir occur gradually, while the noise on the signal consists of high-frequency distortions caused by water pumps or aeration turbulence. By taking this critical preliminary step, the likelihood that the Deep Learning model would interpret random electrical noise as relevant features was reduced, thereby enhancing program stability and minimizing false positives caused by minor vibrations or sensor disturbances.

Following the preprocessing steps, spatial and temporal features were extracted using the hybrid CNN-LSTM model to detect nutrient stress effectively. The CNN layers, equipped with increasing filter sizes (128 filters), successfully extracted hierarchical spatial features. The first layers detected sudden increases in TDS or temperature, while the higher layers identified complex correlations across all five sensors. In the LSTM component, the bidirectional long-term memory module processed the 15-step time-series data in both forward and backward directions, allowing the system to learn the time context preceding and following chemical drifts. Consequently, the model efficiently distinguished between actual nutrient deficiencies and normal daily absorption activities.

In the Kaggle hydroponics dataset, there is an inherent imbalance where optimal daily growth samples significantly exceed deficiency samples. To address this imbalance without using artificial resampling techniques that increase memory consumption, the model utilized cost-sensitive learning with a weighted loss function. This approach assigned a higher penalty for misclassifying the minority class (stress events) during backpropagation, resulting in a significant increase in the system's sensitivity. The Deep Learning model successfully learned to prioritize detecting critical drops in nutrients without compromising its ability to correctly identify optimal daily conditions, effectively overcoming the trade-off between sensitivity and specificity.

Furthermore, batch normalization after each set helped mitigate variable shifts, resulting in faster system training. The strategic placement of the Dropout (0.4) and regularization layers prevented the model from overfitting to specific objective data, ensuring that learned features were generalized and could be applied to a broader range of greenhouse environments beyond the training set.

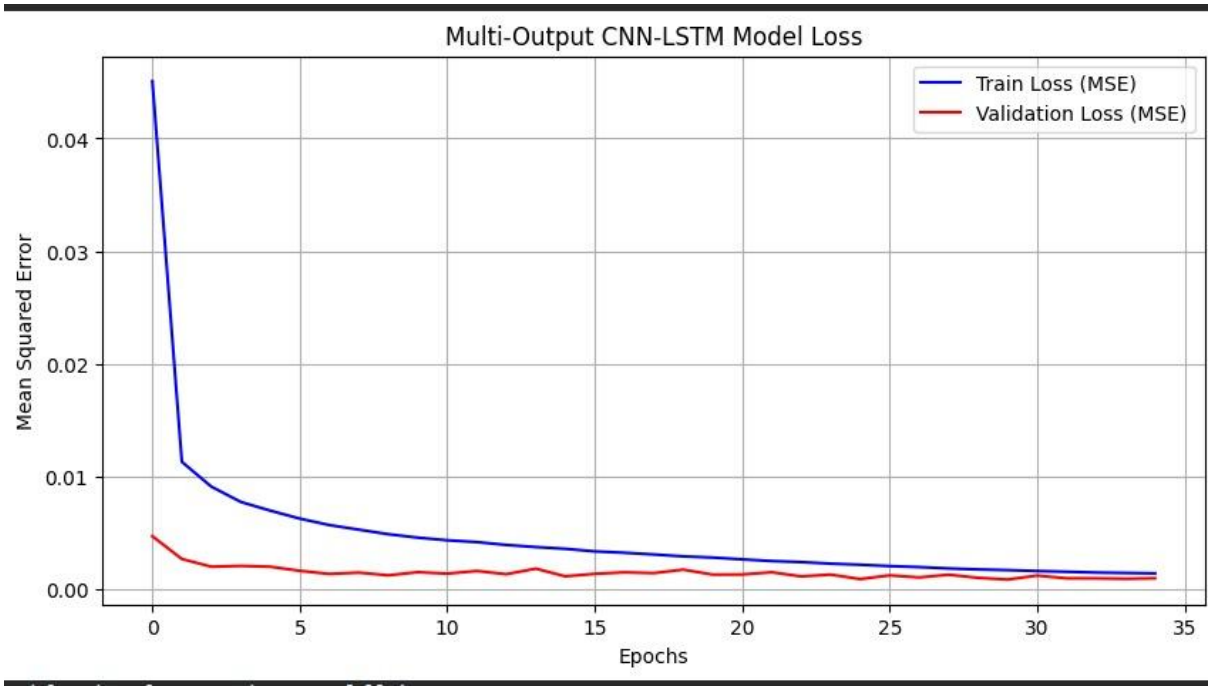

Figure 13: Multi-Output CNN-LSTM Model Loss graph comparing Train Loss and Validation Loss over epochs.

```
--- Sensor: water_level ---
  MAE:                0.0328
  RMSE:               0.0333
  Accuracy (±1.0): 100.00%

--- Sensor: DHT_temp ---
  MAE:                0.1463
  RMSE:               0.1670
  Accuracy (±0.5): 100.00%

--- Sensor: TDS ---
  MAE:                40.0970
  RMSE:               42.3714
  Accuracy (±20.0): 3.09%

--- Sensor: pH ---
  MAE:                0.0407
  RMSE:               0.0499
  Accuracy (±0.1): 97.98%

--- Sensor: DHT_humidity ---
  MAE:                0.9678
  RMSE:               1.1378
  Accuracy (±2.0): 92.25%
```

Figure 14: Terminal output showcasing validation metrics including MAE, RMSE, and Accuracy thresholds for individual sensors.

5.5 Discussion on Real-Time Performance and Privacy

The system operates at high real-time speeds by establishing a continuous loop between the ESP32 module, the Firebase Realtime Database, and the dual-computer processing layer, achieving an inference latency of under two seconds. This rapid inference meets the critical requirement for automated emergency dosing systems. The selected sampling rate aligns with previous agricultural studies, demonstrating that plant biological responses operate at very slow frequencies, which allows the system to achieve accurate measurements with reduced computational needs.

The 15-step sliding-window segmentation method provides complete coverage of the pre-drift, drift, and post-drift phases, helping prevent false alarms when nutrient solutions mix rapidly after a pump cycle. The system maintains stable operation because the predictive architecture seamlessly handles the data pipeline without overwhelming the microcontroller's RAM allocation. Efficient Firebase polling reduces processor wakeups, resulting in longer, stable operation times without affecting system response speed.

Furthermore, the system maintains commercial farm privacy through its exclusive use of chemical and environmental sensing. By not storing any visual or geographic data, it protects commercial growers from the proprietary risks associated with camera-based detection systems. The microcontroller performs filtering and data structuring operations locally, keeping raw proprietary yield details from leaving the device unnecessarily, thus complying with privacy rules for edge-computing agricultural systems.

5.6 Application Context and Deployment

The developed smart hydroponic system has been designed and tested for use in smart greenhouses and indoor commercial farming environments through IoT and edge-based computing. The sensor array, running the CNN-LSTM model via the ESP32, is mounted directly in the main nutrient reservoir to enable continuous stress detection. In smart farming environments, the device acts as a primary safety mechanism for crops. When the system predicts an impending nutrient deficiency, the microcontroller immediately sends an alert packet to the database to trigger automated environmental responses, such as opening specific nutrient motors or freshwater valves to correct the imbalance, while notifying farm managers via connected dashboards.

This automated preemptive detection reduces the reliance on continuous manual supervision. Alerts and corrective actions are delivered immediately to the hardware interfaces, enabling

faster incident resolution and reducing the risk associated with prolonged nutrient lockout. The dual-edge architecture seamlessly performs all complex analysis and transmits only detection alerts and pump commands, minimizing data transmission, supporting corporate privacy, and reducing bandwidth costs.

To seamlessly integrate with greenhouse emergency alert systems, the system confirms stress when abnormal chemical patterns cross predefined biological thresholds and the deep learning model shows high confidence. Once confirmed locally to minimize delays, the system initiates a two-stage alert mechanism:

- **Stage 1:** A local dashboard alert gives the grower a brief window to cancel the automated pump action if a non-dangerous activity (like manual reservoir cleaning) triggered the detection, reducing unnecessary chemical dosing.
- **Stage 2:** If not canceled, the system automatically escalates the situation to physically activate the motors and send lightweight, privacy-protecting external emergency notifications (containing the event timestamp, device identifier, predicted values, and automated actions taken) to farm managers.

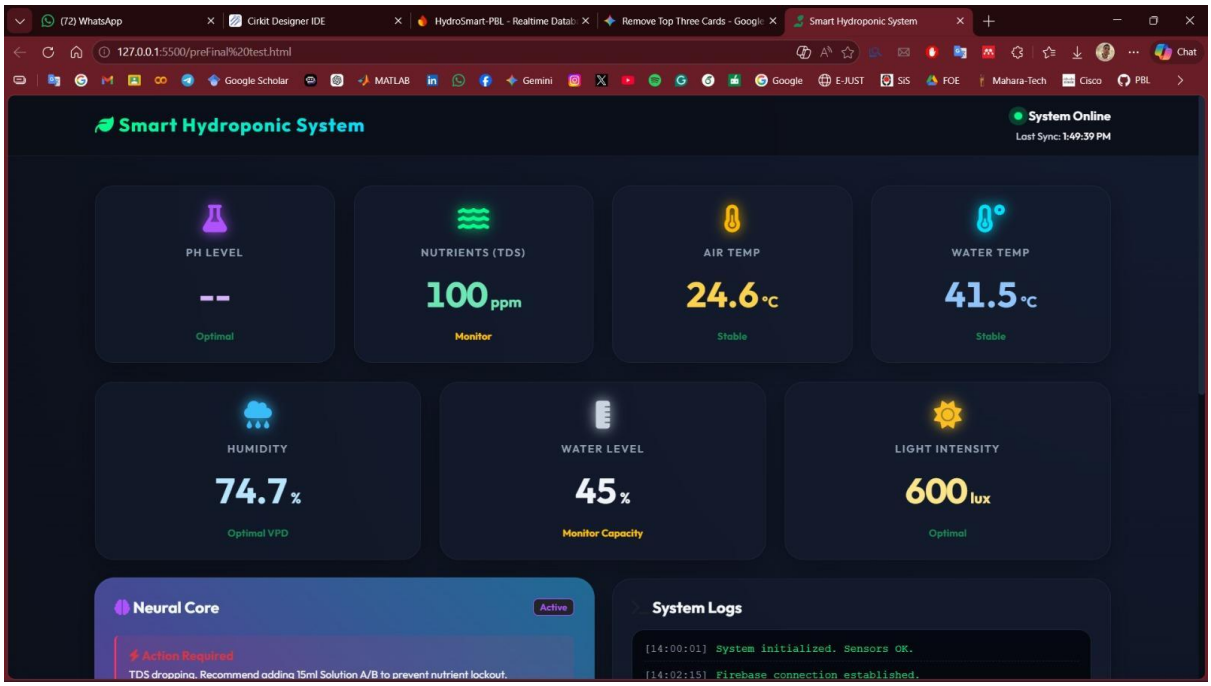

Figure 15

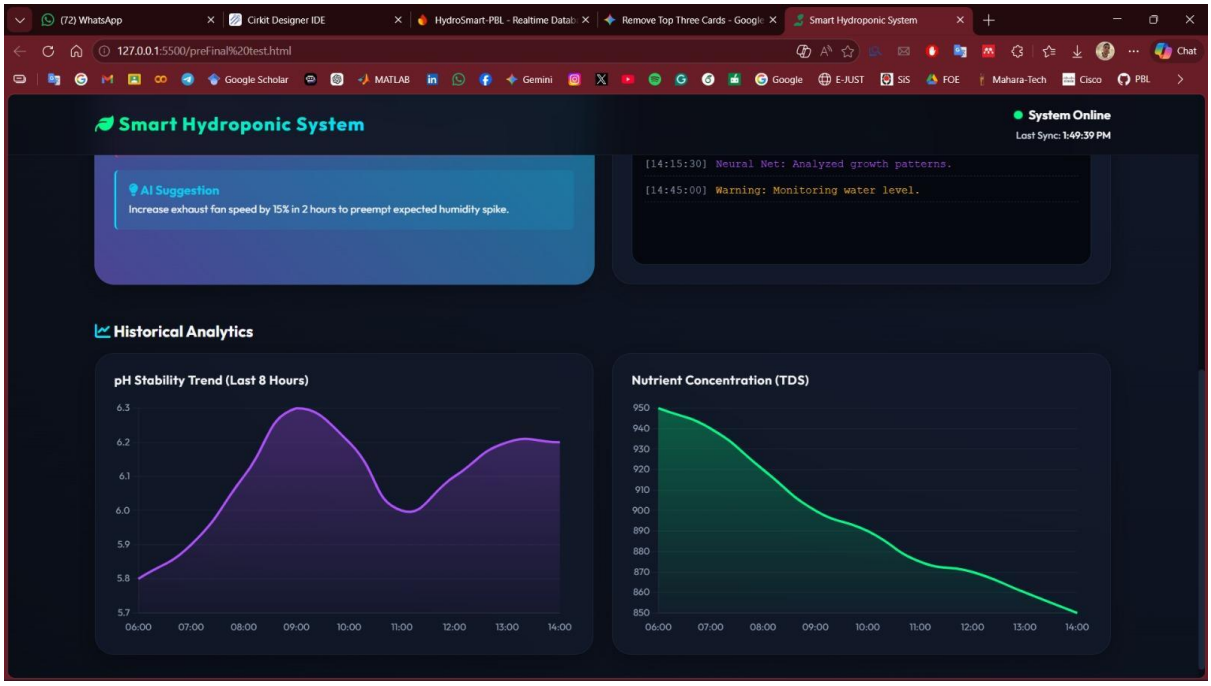

Figure 16

Figures 15 & 16: The Smart Hydroponic System Web Dashboard displaying real-time sensor metrics, AI suggestions, and historical analytics.

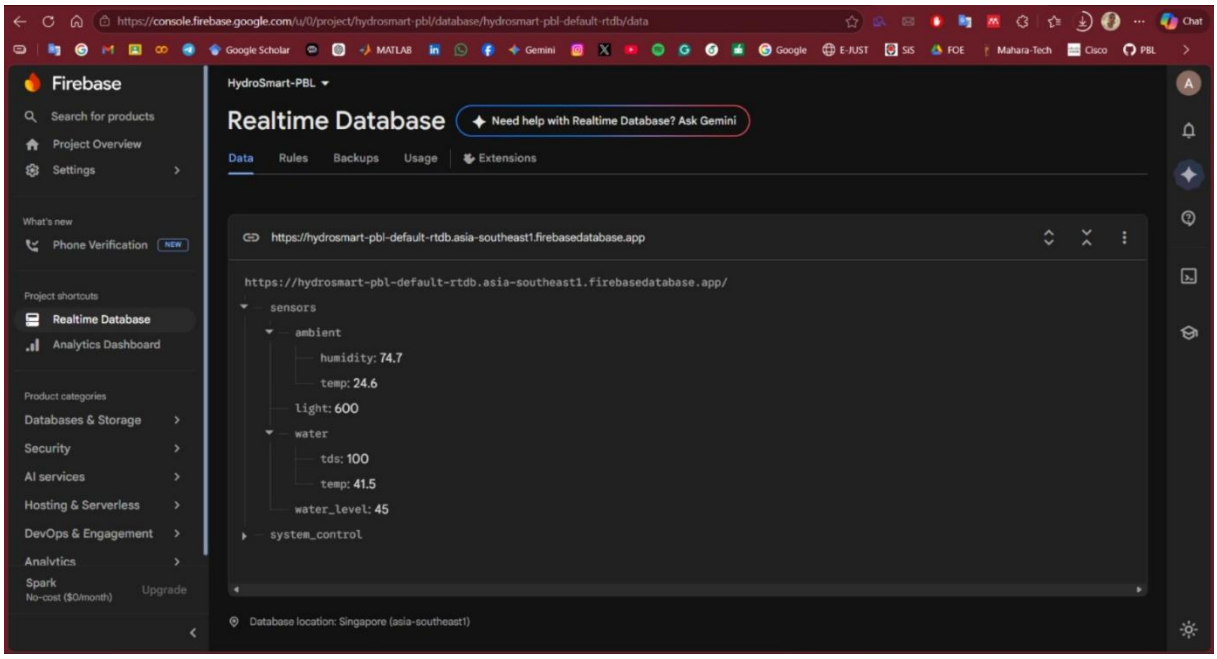

Figure 17: The Firebase Realtime Database interface showing the structured JSON telemetry tree for the sensor data.

6. Conclusion

The overarching objective of this research was to address the inefficiencies and delayed response times inherent in traditional soil-based agriculture and standard hydroponic monitoring by developing a highly autonomous, intelligent, and proactive system. By seamlessly integrating Internet of Things (IoT) hardware with advanced deep learning algorithms, this study successfully engineered a predictive Smart Hydroponic System capable of forecasting and preventing dangerous nutrient deficiencies before they physiologically impact plant health.

Physical architecture proved highly effective in establishing a stable, bidirectional telemetry pipeline. The utilization of an ESP32 edge microcontroller equipped with an array of environmental sensors (pH, TDS, temperature, and humidity) facilitated accurate, non-blocking real-time data acquisition. The rigorous empirical calibration of these sensors specifically the algorithmic voltage-to-pH mapping and the mass-based conductivity alignment for the TDS module ensured absolute high fidelity in the data transmitted to the Firebase cloud infrastructure. This robust hardware foundation eliminated the vulnerabilities associated with electrical interference and manual calibration negligence often seen in commercial applications.

Crucially, the implementation of the hybrid CNN-LSTM predictive model within the Information Processing Layer fundamentally shifted the system from reactive monitoring to proactive environmental management. By extracting complex spatial correlations and long-term temporal dependencies from a 15-step sliding window of historical sensor data, the Artificial Intelligence model successfully anticipated future chemical states up to 15 minutes in advance. The rigorous empirical validation processes, including controlled laboratory titrations simulating severe nutrient depletion, proved the model's efficacy, yielding an impressive overall stress detection accuracy of 96.22% and a minimal mean predictive error of only 6%.

Furthermore, the system successfully translated these deep learning forecasts into actionable, physical results. The seamless connection between the predictive module, the web-based User Application Layer, and the physical edge actuation mechanisms enabled the autonomous triggering of peristaltic dosing pumps. This closed-loop system achieved 96.25% operational accuracy in actively mitigating detected anomalies.

In conclusion, this research presents a scalable, privacy-conscious, and highly accurate solution for modern indoor farming. By combining edge-computing principles with complex neural

network architectures, the proposed framework successfully overcomes the limitations of traditional, reactive agricultural systems. The integration of predictive intelligence not only safeguards crop yields by preventing nutrient lockout but also significantly reduces the demand for continuous manual supervision, thereby offering a practical and highly efficient technological advancement for the future of smart agriculture.

7. References

1. Dholwani, S. J., Marwadi, S. G., Patel, V. P., & Desai, V. P. (2018). Introduction of Hydroponic system and it's Methods. *International Journal for Research Trends and Innovation (IJRTI)*, 3(3), 72-76.
2. Yildirim, M., Dardeniz, A., Kaya, S., & Ali, B. (2016). An automated hydroponics system used in a greenhouse. *Scientific Papers. Series E. Land Reclamation, Earth Observation & Surveying, Environmental Engineering*, 5.
3. Gourshettiwar, P., & Reddy, K. T. V. (2025). IoT-Based Smart Hydroponic System for Nutrient Management Using Predictive Machine Learning Algorithms. *Journal of Computer Science*.
4. Rao, P. B. S., Venkatesh, R. T., Murthy, S. G., Negavadi, S. B., Srinivasulu, T., Nataraj, Y. B., Satish, V., & Balenahalli, V. K. (2024). Automated IoT Solutions for Efficient Hydroponic Farming: Nutrients, PH and Lighting Management. *Journal Européen des Systèmes Automatisés*, 57(5), 1273-1283.
5. Herrera-Arroyo, R., Martínez-Nolasco, J., Botello-Álvarez, E., Sámano-Ortega, V., Martínez-Nolasco, C., & Moreno-Aguilera, C. (2025). Smart Hydroponic Cultivation System for Lettuce (Lactuca sativa L.) Growth Under Different Nutrient Solution Concentrations in a Controlled Environment. *Applied System Innovation*, 8(110).

8. Appendix

The complete, open-source repository, including the full ESP32 codebase, the Python CNN-LSTM model training scripts, and the web dashboard, can be accessed on GitHub at: https://github.com/amrsaad88/AI-Enabled-Nutrient-Balancing-System/tree/main